Kyler Li P. Tolentino

FA6

Link of Google Colab using R programming (All questions): https://colab.research.google.com/drive/1KDXb_1Hcqg7Esnm3sWg8k8_aSVoNr9lz?usp=sharing

SOLUTION (Handwritten):

8.21

a.) Population Mean

$$\mu = \frac{3+7+11+15}{4} = \boxed{9}$$

b.)
$$\sigma^2 = \frac{(3-9)^2+(7-9)^2+(11-9)^2+(15-9)^2}{4}$$
$$= 20 \qquad \sigma = \sqrt{20} \approx \boxed{4.47}$$

c.) $\mu_{\bar{x}} = \mu = \boxed{9}$

d.) $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{\sqrt{20}}{\sqrt{2}} = \sqrt{10} \approx \boxed{3.16}$

8.34

$n = 200$, $p = 0.5$, $q = 1-p = 0.5$

Mean: $\mu_{\hat{p}} = p = 0.5$

$$\sigma_{\hat{p}} = \sqrt{\frac{p \cdot q}{n}} = \sqrt{\frac{(0.5)(0.5)}{200}} = \sqrt{0.00125}$$
$$= 0.035355$$

a.) $z = \frac{0.40-0.50}{0.035355} \approx -2.83 \approx \boxed{0.0023}$

b.) $z_1 = \frac{0.43-0.50}{0.035355} \approx -1.98$ $\qquad 0.9761 -$

$z_2 = \frac{0.57-0.50}{0.035355} \approx 1.98$ $\qquad 0.0239$

$= \boxed{0.9522}$

c.) $z = \frac{0.54-0.50}{0.035355}$

$1 - 0.8708 = \boxed{0.1292}$

Sterling

8.49

a.) Population Mean

$\mu = \sum[x \cdot P(x)]$

$\mu = (6)(0.1) + (9)(0.2) + (12)(0.4) + (15)(0.2) + (18)(0.1)$

$\mu = \boxed{12}$

b.) Population variance

$\sigma^2 = \sum[x^2 \cdot P(x)] - \mu^2$

$= (6^2)(0.1) + (9^2)(0.2) + (12^2)(0.4) + (15^2)(0.2) + (18^2)(0.1)$

$= 154.8$

$= 154.8 - (12)^2$

$= \boxed{10.8}$

c.)

$\frac{x_1 + x_2}{2}$ easy in R